

Tata Steel Equipment Maintenance Manual

Engineering & Maintenance Department | 2024

Section 1: Maintenance Philosophy

Tata Steel follows a Total Productive Maintenance (TPM) approach combining Preventive, Predictive, and Corrective maintenance strategies to maximize equipment availability while minimizing unplanned downtime and cost.

****Maintenance Goals:****

- Equipment availability above 95% for critical assets
- Reduce unplanned downtime by 20% year-on-year
- Mean Time Between Failures (MTBF) improvement
- Mean Time To Repair (MTTR) reduction

Section 2: Maintenance Schedule Types

****Preventive Maintenance (PM):****

- Scheduled at fixed intervals based on manufacturer recommendations and operating history
- Daily, weekly, monthly, quarterly, and annual checks
- Prevents unexpected breakdowns through planned servicing
- Logged in Computerized Maintenance Management System (CMMS)
- Includes lubrication, cleaning, calibration, and component replacement on schedule

****Predictive Maintenance (PdM):****

- Based on continuous sensor data monitoring (temperature, vibration, pressure, current)
- Condition-based triggers rather than fixed schedules
- Maintenance triggered when readings cross warning thresholds
- Reduces unnecessary maintenance on healthy equipment

- Techniques used: vibration analysis, thermography, oil analysis, ultrasonic testing

****Corrective Maintenance (CM):****

- Performed after failure occurs (reactive)
- Emergency response within 2 hours for critical equipment
- Root cause analysis (RCA) mandatory for repeat failures
- Failure report filed within 24 hours on CMMS
- RCA techniques: 5-Why analysis, Fishbone diagram

****Shutdown Maintenance:****

- Planned annual/bi-annual shutdowns for major overhauls
- Requires detailed work order planning 3 months in advance
- Critical spares procurement initiated 6 months ahead

Section 3: Equipment Codes & Standards

Equipment	Code	PM Frequency	Critical Level	MTBF Target
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Crusher Machine	EQ001	Weekly	High	30 days
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Conveyor Belt	EQ002	Daily	High	45 days
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Heating Regulator	EQ003	Monthly	Medium	60 days
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Control Room Systems	EQ004	Monthly	Critical	90 days
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Locomotive	EQ005	Weekly	High	30 days
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Blast Furnace	EQ006	Daily	Critical	365 days
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Battery (Coke Oven)	EQ007	Daily	Critical	180 days
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Stacker/Reclaimer	EQ008	Weekly	Medium	45 days
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Section 4: Troubleshooting Guide

****Crusher Machine (EQ001):****

Problem: Abnormal vibration

Possible causes:

1. Worn out bearings
2. Loose bolts/foundation
3. Material blockage in crushing chamber
4. Belt or coupling misalignment
5. Rotor imbalance

Immediate steps:

1. Stop machine immediately
2. Lock out power supply (LOTO procedure)
3. Check for visible blockage and clear if safe
4. Inspect bearing condition (heat, noise, play)
5. Check foundation bolts for looseness
6. Call maintenance team if unresolved within 30 minutes

Problem: Overheating

Possible causes:

1. Insufficient lubrication
2. Overloading beyond rated capacity
3. Cooling system failure
4. Bearing degradation

Steps:

1. Reduce load immediately
2. Check lubrication levels and quality
3. Inspect cooling system (fans, water circulation)
4. Allow to cool 30 minutes before restart
5. If recurring, schedule bearing inspection

Problem: Reduced output / capacity drop

Possible causes:

1. Worn crushing plates/jaws
2. Incorrect feed size
3. Belt slippage on drive

Steps:

1. Check feed material size against spec
2. Inspect crushing plate wear (measure against tolerance)
3. Check drive belt tension

****Conveyor Belt (EQ002):****

Problem: Belt slipping

Steps:

1. Check belt tension against specification
2. Inspect drive pulley for wear/lagging damage
3. Check for material spillage causing buildup
4. Adjust tension if needed using takeup mechanism
5. Check for moisture/contamination on pulley surface

Problem: Unusual noise

Steps:

1. Identify noise source (idler, pulley, or belt)
2. Check all rollers/idlers for bearing wear
3. Inspect belt for damage (tears, splice failure)
4. Check lubrication of bearings
5. Verify belt tracking/alignment

Problem: Belt tracking off-center

Steps:

1. Check idler alignment along entire conveyor length
2. Inspect for uneven loading causing belt drift
3. Adjust training idlers
4. Check structure for tilting/settling

****Heating Regulator (EQ003):****

Problem: Temperature fluctuation beyond setpoint

Steps:

1. Check fuel supply pressure and flow
2. Inspect thermocouple/sensor calibration
3. Verify control valve operation
4. Check for air leaks in combustion chamber
5. Review control logic settings in SCADA

Problem: Error code displayed

Steps:

1. Note exact error code and timestamp
2. Refer to equipment-specific error code manual
3. Check sensor wiring connections
4. Restart controller if safe (follow restart procedure)
5. Escalate to control system team if error persists

****Locomotive (EQ005):****

Problem: Brake system warning

Steps:

1. Stop locomotive in safe location
2. Do not proceed until brake system inspected
3. Check air pressure gauges
4. Inspect brake pads/shoes for wear

5. Contact rolling stock maintenance team

Section 5: Sensor Thresholds

Sensor	Normal Range	Warning	Critical
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Temperature	20-80°C	81-95°C	Above 95°C
Vibration	0-0.5 mm/s	0.5-1.0 mm/s	Above 1.0 mm/s
Pressure	100-130 PSI	131-150 PSI	Above 150 PSI
RPM	800-1200	1201-1400	Above 1400
Oil Temperature	40-70°C	71-85°C	Above 85°C
Current Draw	Rated ±10%	Rated ±20%	Rated ±30%

Action Matrix:

Status	Action Required	Timeline
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Normal	Continue routine monitoring	-
Warning	Schedule inspection, increase monitoring frequency	Within 48 hours
Critical	Stop equipment, immediate maintenance intervention	Immediate

Section 6: Spare Parts Management

- Critical spares maintained at minimum stock levels defined per equipment
- Reorder triggered automatically when stock falls below reorder point
- Long lead-time items (bearings, motors, control cards) reviewed quarterly
- Obsolete equipment spares cross-referenced for alternate suppliers

Spare Category	Examples	Min Stock
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| Bearings | Crusher, conveyor bearings | 4 units |

| Belts | Conveyor belts, V-belts | 2 units |

| Sensors | Temperature, vibration probes | 6 units |

| Electrical | Contactors, relays, fuses | 10 units |

Section 7: Permit to Work System

All maintenance work requires appropriate permit:

| Permit Type | Required For |

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| Hot Work Permit | Welding, cutting, grinding |

| Confined Space Permit | Tank, vessel, pit entry |

| Electrical Work Permit | Any electrical isolation/work |

| Excavation Permit | Digging near underground services |

| Working at Height Permit | Work above 1.8m |

Section 8: Emergency Maintenance Contacts

| Role | Contact |

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| Maintenance Supervisor | Extension 301 |

| Electrical Team | Extension 302 |

| Mechanical Team | Extension 303 |

| Control Room | Extension 304 |

| Instrumentation Team | Extension 305 |

| Emergency Response | Extension 999 |